Table C3. Fully unconstrained random parameters binary logit models with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), estimated separately for each year: single-vehicle crashes without a motorcycle, Thailand, 2016–2023. Estimated parameter (*t*-statistic).

Variable description	2016	2017	2018	2019	2020	2021	2022	2023
Constant [serious injury / fatal]								
Constant	–1.427 (–14.56)	–2.171 (–14.18)	–1.824 (–12.00)	–2.369 (–16.91)	–2.475 (–16.52)	–2.854 (–17.96)	–3.000 (–18.22)	–3.125 (–18.58)
Vehicle involvement								
One passenger car involved (1 if yes, 0 otherwise)		0.163 (1.93)		0.198 (2.31)	0.261 (3.21)	0.153 (1.83)	–0.149 (–0.61)	
(Standard deviation of parameter distribution)							0.982 (2.50)	
One 6-wheel truck involved (1 if yes, 0 otherwise)	–0.526 (–2.91)		–0.366 (–2.05)	–0.518 (–2.50)			–0.374 (–1.87)	
One 10-wheel truck involved (1 if yes, 0 otherwise)	–0.889 (–4.08)	–0.404 (–1.94)	–1.067 (–4.27)	–0.603 (–2.58)	–0.452 (–2.04)	–0.729 (–3.00)	–0.664 (–2.66)	
One truck with more than 10 wheels involved (1 if yes, 0 otherwise)	–0.828 (–5.54)	–0.790 (–4.90)	–1.035 (–6.54)	–1.147 (–5.71)	–0.777 (–4.52)	–0.515 (–3.61)	–0.664 (–4.24)	–0.550 (–3.38)
Van involved (1 if yes, 0 otherwise)	0.680 (3.46)	0.999 (4.52)	0.516 (2.50)	0.876 (4.20)		0.979 (3.52)	0.594 (2.28)	0.622 (2.12)
Pedestrian involved (1 if yes, 0 otherwise)	4.054 (10.98)	4.458 (10.90)	3.560 (11.12)	4.232 (13.95)	4.135 (15.04)	4.409 (13.24)	3.984 (13.22)	4.792 (15.41)
Crash type								
Rear-end collision in the same lane (1 if yes, 0 otherwise)	1.004 (4.20)	0.654 (2.33)		0.994 (2.74)	1.475 (4.54)	1.186 (3.29)	1.305 (4.12)	1.177 (3.51)
Ran off road to the left and hit an object (1 if yes, 0 otherwise)	–0.626 (–5.01)	–0.805 (–5.57)	–0.734 (–5.60)					–4.390 (–2.15)
(Standard deviation of parameter distribution)								4.507 (2.83)
Vehicle mounted a traffic island (1 if yes, 0 otherwise)	–0.751 (–5.48)	–0.766 (–5.57)	–1.047 (–7.92)	–0.735 (–4.14)	–0.527 (–4.06)	–0.369 (–2.93)	–0.462 (–3.74)	–5.659 (–1.84)
(Standard deviation of parameter distribution)								4.446 (2.21)
Ran off road to the right and hit an object (1 if yes, 0 otherwise)	–0.581 (–4.39)	–0.638 (–4.38)	–0.728 (–5.07)					–0.498 (–3.78)
Ran off road to the left (1 if yes, 0 otherwise)	0.457 (3.61)	0.424 (3.16)	0.273 (1.99)	0.600 (5.02)	0.600 (5.15)	0.766 (6.16)	1.087 (8.60)	0.763 (5.39)
Ran off road on a right bend and hit an object (1 if yes, 0 otherwise)	–0.751 (–4.59)	–0.697 (–4.21)	–0.833 (–5.02)					–0.335 (–2.04)
Ran off road on a left bend and hit an object (1 if yes, 0 otherwise)	–0.817 (–4.56)	–0.414 (–2.50)	–0.860 (–4.96)		–0.792 (–3.02)			
Ran off road to the right (1 if yes, 0 otherwise)	0.616 (4.37)	0.640 (4.34)		0.609 (4.46)	0.594 (4.30)	0.785 (5.00)	0.734 (4.50)	0.560 (3.34)
Ran off road on a left bend (1 if yes, 0 otherwise)			–0.593 (–2.81)	1.378 (4.29)		0.464 (2.29)		0.398 (1.69)
Ran off road on a right bend (1 if yes, 0 otherwise)			–0.523 (–2.42)	0.717 (2.22)		0.933 (5.35)	0.793 (3.60)	0.473 (1.96)
Crash cause								
Speeding (1 if yes, 0 otherwise)		0.303 (2.32)	0.250 (1.98)	0.257 (1.93)	0.441 (3.27)	0.574 (3.75)	0.663 (4.10)	0.624 (4.47)
Cutting in too closely (1 if yes, 0 otherwise)						0.590 (1.76)	0.974 (3.05)	0.868 (2.23)
Driver fell asleep (1 if yes, 0 otherwise)	0.567 (5.14)	0.754 (4.62)	0.658 (4.32)	0.711 (4.57)	0.887 (5.58)	0.582 (3.07)	0.742 (3.81)	1.035 (6.11)
Roadway characteristics								
Crash on the main carriageway (1 if yes, 0 otherwise)	–0.698 (–3.73)	–0.466 (–2.58)	–0.696 (–3.75)		–0.354 (–1.84)		–0.465 (–2.06)	
Road under construction (1 if yes, 0 otherwise)				0.520 (1.89)		0.471 (1.90)		
Painted median (1 if yes, 0 otherwise)		–0.423 (–2.20)			–0.424 (–2.46)			
Raised or earth median (1 if yes, 0 otherwise)	–0.260 (–2.47)	–0.282 (–2.69)	–0.188 (–1.67)	–0.443 (–1.56)				
(Standard deviation of parameter distribution)				1.335 (3.31)				
Depressed median (1 if yes, 0 otherwise)	–0.149 (–1.68)		0.156 (1.67)		–0.190 (–2.18)			
Barrier median (1 if yes, 0 otherwise)	–0.644 (–4.23)	–2.396 (–2.66)	–0.399 (–2.66)	–0.905 (–5.90)	–0.899 (–6.36)	–1.086 (–7.92)	–0.883 (–6.72)	–1.012 (–6.43)
(Standard deviation of parameter distribution)		2.455 (3.11)						
Horizontal curve (1 if yes, 0 otherwise)				–2.037 (–2.73)	–1.166 (–2.18)			
(Standard deviation of parameter distribution)				2.952 (3.91)	2.312 (3.90)			

Variable description	2016	2017	2018	2019	2020	2021	2022	2023
Concrete pavement surface (1 if yes, 0 otherwise)						–1.634 (–2.48)		
(Standard deviation of parameter distribution)						1.902 (3.14)		
Sag vertical curve (1 if yes, 0 otherwise)	0.445 (1.92)		0.463 (1.72)					
T-intersection (1 if yes, 0 otherwise)	–0.501 (–1.83)				–0.981 (–2.66)			–0.633 (–2.20)
U-turn with deceleration lane only (1 if yes, 0 otherwise)				0.664 (2.30)	–0.527 (–1.79)			
Bridge section (1 if yes, 0 otherwise)		0.728 (3.10)	0.609 (2.52)	0.607 (2.36)		0.772 (3.28)	0.760 (2.75)	0.533 (1.76)
Environmental characteristics								
Wet road surface (1 if yes, 0 otherwise)			–0.223 (–2.16)	–0.250 (–1.97)	–1.426 (–2.57)	–0.522 (–5.09)		–0.369 (–3.05)
(Standard deviation of parameter distribution)					1.726 (3.03)			
Dark, no street lighting (1 if yes, 0 otherwise)	0.558 (5.20)	0.379 (3.04)	0.591 (5.04)	0.728 (5.99)	0.502 (4.26)	0.495 (3.94)	0.382 (2.99)	
Dark, with street lighting (1 if yes, 0 otherwise)						0.172 (2.15)		
Temporal characteristics								
Weekend (1 if yes, 0 otherwise)		–0.510 (–1.83)						0.255 (2.83)
(Standard deviation of parameter distribution)		1.473 (4.03)						
Songkran festival period (1 if yes, 0 otherwise)	–1.030 (–2.27)					–1.223 (–1.87)		
(Standard deviation of parameter distribution)	1.682 (2.77)					2.071 (2.91)		
New Year festival period (1 if yes, 0 otherwise)	–0.546 (–1.57)		–0.525 (–1.44)		0.256 (1.67)		0.286 (1.71)	
(Standard deviation of parameter distribution)	1.837 (3.22)		1.803 (3.34)					
Evening (1 if yes, 0 otherwise)		0.254 (2.19)	0.176 (1.73)					0.267 (2.44)
Midnight / early morning (1 if yes, 0 otherwise)	0.199 (2.52)	0.287 (2.72)	0.272 (2.98)					0.170 (1.69)
Peak hours (1 if yes, 0 otherwise)		0.202 (1.74)						
Region								
Eastern region (1 if yes, 0 otherwise)						0.286 (2.40)		0.465 (3.17)
Northeastern region (1 if yes, 0 otherwise)					0.195 (2.16)			0.324 (2.70)
Northern region (1 if yes, 0 otherwise)			–0.764 (–1.55)			–0.219 (–1.84)	–2.299 (–2.55)	
(Standard deviation of parameter distribution)			2.251 (3.68)				2.344 (3.15)	
Southern region (1 if yes, 0 otherwise)		0.226 (2.35)	0.255 (2.65)					0.582 (4.79)
Western region (1 if yes, 0 otherwise)			0.468 (3.98)	0.608 (4.89)	0.773 (6.50)	0.375 (3.00)		0.569 (3.65)
Heterogeneity in the means of random parameters								
Ran off road to the left and hit an object: Dark, no street lighting								1.540 (2.00)
Horizontal curve: Eastern region				1.189 (2.89)	1.021 (2.57)			
Horizontal curve: Northern region				–0.754 (–2.23)				
Concrete pavement surface: Ran off road on a left bend and hit an object						1.428 (2.66)		
Concrete pavement surface: Southern region						1.464 (3.02)		
Wet road surface: Road under construction					2.076 (2.38)			
Weekend: Depressed median		0.330 (1.84)						
Songkran festival period: Dark, with street lighting	1.429 (3.28)							
New Year festival period: Northern region	–1.417 (–2.60)							

Variable description	2016	2017	2018	2019	2020	2021	2022	2023
Northern region: Ran off road to the right and hit an object							1.097 (2.41)	
Northern region: Concrete pavement surface			–1.550 (–1.97)					
Northern region: Raining			–0.810 (–1.99)					
Northern region: Evening							0.885 (2.32)	
Heterogeneity in the variances of random parameters								
Ran off road to the left and hit an object: One 6-wheel truck involved								–0.296 (–1.83)
Horizontal curve: Raining				–0.268 (–2.16)				
Wet road surface: Northern region					0.347 (1.70)			
Songkran festival period: Peak hours	0.914 (2.11)							
Model statistics								
Number of observations	7,482	7,911	8,693	8,808	9,667	9,882	10,320	10,968
Number of estimated parameters	30	30	34	29	29	30	25	31
Random parameters / heterogeneity in means / in variances	2 / 2 / 1	2 / 1 / 0	2 / 2 / 0	2 / 2 / 1	2 / 2 / 1	2 / 2 / 0	2 / 2 / 0	2 / 1 / 1
Log-likelihood at zero, LL(0)	–5186.13	–5483.49	–6025.53	–6105.24	–6700.65	–6849.68	–7153.28	–7602.44
Log-likelihood with constant only, LL(C)	–3261.07	–3151.43	–3354.88	–3412.71	–3610.25	–3262.28	–3196.08	–3405.09
Log-likelihood at convergence, LL(β)	–2853.52	–2768.24	–2959.00	–3021.74	–3203.59	–2907.88	–2857.56	–2984.43
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$	0.450	0.495	0.509	0.505	0.522	0.575	0.601	0.607

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median, one pickup truck. Each column is a separate model for one year. Each model allows at most two random parameters, two heterogeneity-in-means terms and one heterogeneity-in-variance term, searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level. A blank cell means the variable is not in that year's model. Rows with a standard deviation give the mean of the random parameter in the years where it is random and a fixed parameter in the other years. A variable used in a heterogeneity function enters only the random parameter shown in that year and is removed from the main utility. Random parameters normally distributed; 2,000 Halton draws.